

Question C.1 — Deterministic Hierarchy of Swap Times

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All numerical measurements in this section come from the instrumented campaign of 2 000 runs (20 amplitudes $\times$ 100 seeds, $H = 2\,000$, baseline $\hat{p}_0$ estimated over the last 3 000 pre-drift steps) documented in `04_experimentations/resultats_R2/`.

1 Deterministic Relation Between τ_{ARF} and $\tau_{\text{swap}}(q)$

Let N_t be the number of distinct trees replaced at time t since the drift instant τ^* , and

$$\tau_{\text{swap}}(q) = \inf\{t > \tau^* : N_t/M \geq q\} - \tau^*.$$

N_t can only increase (a replaced tree stays counted), so τ_{swap} is increasing in q . And since N_t is an integer, the condition $N_t \geq 1$ is equivalent to $N_t/M \geq 1/M$: τ_{ARF} is therefore exactly $\tau_{\text{swap}}(1/M)$, i.e. $\tau_{\text{swap}}(0.10)$ for $M = 10$. Hence, for every $q \geq 1/M$:

$$\boxed{\tau_{\text{ARF}} \leq \tau_{\text{swap}}(q)} \quad \text{on every run, with no probabilistic assumption.} \quad (1)$$

Quotas actually achieved. Since N_t only takes integer values, a quota q is only ever met once N_t reaches $\lceil qM \rceil$. With $M = 10$, this lands on an exact tree count for $q = 10\%$ (1 tree) and $q = 50\%$ (5 trees), but *not* for $q = 25\%$ or $q = 75\%$: the labels $\tau_{\text{swap}}(25\%)$ and $\tau_{\text{swap}}(75\%)$ used throughout this section actually correspond to the first integer counts ≥ 2.5 and ≥ 7.5 , i.e. 3 trees (a true fraction of 30%) and 8 trees (80%) respectively. The 25%/75% labels are kept for consistency with the quotas fixed for the campaign (also used in Q D), but should be read as “at least 3 (resp. 8) trees renewed”, not a literal quarter or three-quarters of the forest.

Pipeline consistency check. Relation (1) is deterministic, so no run can violate it: measuring it on the campaign says nothing about the forest, it checks the instrumentation. Two figures are reported in that light and in no other. $\tau_{\text{ARF}} = \tau_{\text{swap}}(10\%)$ holds on all 2 000 runs with zero gap — necessarily, since $qM = 1$ makes the two the same extraction from the same event list. And no violation of (1) appears across the 5 795 uncensored comparisons for $q \in \{25\%, 50\%, 75\%\}$ — necessarily again, the event list being sorted and N_t non-decreasing. Either result failing would have located a bug in the offline reconstruction, not a property of the ensemble.

What the runs actually show. The sign of the gap being settled in advance, what remains to be measured is its *size* and its *spread*. Figure 1 gives both, run by run rather than through an aggregate: every point lies on or above the equality line by construction, but how far above varies enormously, and the dispersion — invisible in a median alone — grows sharply with q , together with the fraction of runs that never reach the quota at all (red triangles). That dispersion is the empirical content of this section; the next one quantifies it.

Direction of the bias. Since inequality (1) is strict in the vast majority of cases, using τ_{ARF} as the forest’s adaptation date *systematically and one-sidedly underestimates* the time needed for a genuinely significant fraction of the ensemble to have been renewed: a one-sided, optimistic bias that averaging over seeds does not correct. The inequality holding run by run, the error always falls on the same side, so nothing cancels: 10 000 runs would leave the mean just as far below the truth as 2 000 do.

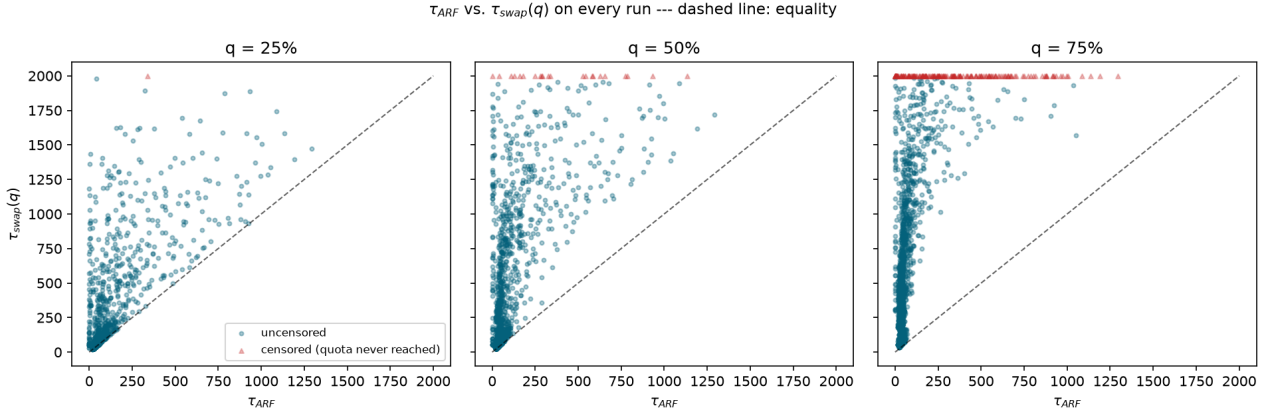


Figure 1: τ_{ARF} versus $\tau_{\text{swap}}(q)$ on every one of the 2000 runs, for $q \in \{25\%, 50\%, 75\%\}$. Dashed line: equality. No point ever falls below it, confirming inequality (1) run by run rather than only in aggregate; red triangles mark runs where the quota was never reached before H .

Not the same bias as Q B.2. Two results in this project conclude that τ_{ARF} is optimistic, and a reader meeting them in sequence will take the second for a confirmation of the first. They share nothing but their direction.

Inequality (1) is *combinatorial*. It is a property of the counting process N_t , it holds pathwise on every run, and it would hold unchanged if the trees were independent. Q B.2 is *probabilistic*. It compares two candidate laws for τ_{ARF} — the article’s independence assumption against the true, positively dependent one — and says nothing whatever about $\tau_{\text{swap}}(q)$.

So they stack. Grant Q B.2’s correction in full, so that the law of τ_{ARF} is known exactly: it still dates the forest’s recovery too early, by the margins measured in the next section. Two errors, compounding; neither confirms the other.

2 Gap $\tau_{\text{swap}}(q) - \tau_{\text{ARF}}$ as a Function of Δe

Inequality (1) only tells us the sign of the gap, not its magnitude. Figure 2 (full campaign, inter-seed median) plots $\tau_{\text{swap}}(q) - \tau_{\text{ARF}}$ as a function of Δe , for $q \in \{10\%, 25\%, 50\%, 75\%\}$. The $q = 10\%$ curve is shown as a trivial zero-baseline reference, not a substantive result: it is flat at 0 everywhere by construction, since $\tau_{\text{ARF}} = \tau_{\text{swap}}(0.10)$.

At the bottom of the grid, the quota is reached by noise. Censoring at $q = 75\%$ is *not* monotone in Δe : 35% at $\Delta e = 0.028$, 78% at 0.085, 54% at 0.141, 13% at 0.194, and 0% from 0.287 upwards. The weakest amplitude on the grid is thus *less* censored than the next one up, which no account in terms of signal strength can explain.

A no-drift control settles it. With no drift injected at all, the forest still renews a median of 9 trees out of 10 within the same horizon, on 14 swap events per run, and only 18% of runs fail to reach the 75% quota. Tree replacement is therefore not evidence of adaptation: ADWIN fires on noise at a substantial rate regardless. At $\Delta e = 0.028$ the campaign records a median of 12.5 swaps per run against 14 under no drift — the drift adds essentially nothing, and the quota fills by accident. The count then *drops* to 8 at $\Delta e = 0.085$, below the no-drift rate, which is where censoring peaks.

Two consequences. First, $\tau_{\text{swap}}(q)$ measures adaptation only where swaps are drift-driven, so the two lowest rows of any table in this section carry a different object from the rest and should not be read as its continuation. Second, the dip below the no-drift rate has no established mechanism here — a plausible one is that an early drift-driven replacement stabilizes the ensemble error and thereby suppresses the noise-driven swaps that would otherwise have followed — and it is reported as a measurement, not as an explanation.

Substitutability criterion. A gap exceeding 100 steps (5% of the horizon $H = 2000$) is deemed disqualifying for τ_{ARF} as a substitute for $\tau_{\text{swap}}(q)$. We claim no external grounding for that figure: no reference fixes what counts as too large a gap here, so this is a declared convention, adopted before the measurements were read.

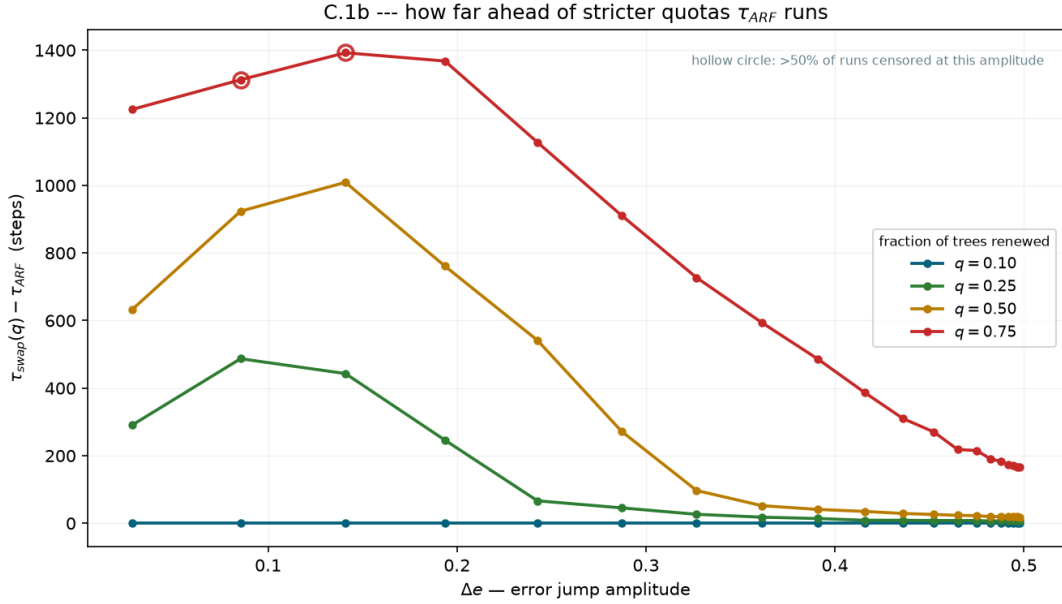


Figure 2: Median gap $\tau_{\text{swap}}(q) - \tau_{\text{ARF}}$ as a function of Δe , full campaign (2000 runs). Hollow circle markers: amplitudes where more than 50% of runs are censored for that quota, so that the median is computed on a minority of runs and is not interpretable. This affects $q = 75\%$ only, at $\Delta e = 0.085$ (78% censored) and $\Delta e = 0.141$ (54%).

What makes it usable is that the conclusions do not turn on it. Re-running the criterion at 50 and at 150 steps moves the amplitude at which τ_{ARF} becomes acceptable again by a single grid step — for $q = 25\%$, from $\Delta e \approx 0.243$ to 0.287 ; for $q = 50\%$, from 0.327 to 0.391 — and leaves the verdict on $q = 75\%$ untouched, the median gap there never falling below 167 steps anywhere on the grid. Any threshold in $[50, 167]$ yields the same three conclusions.

A second criterion, answering a different question. A more operational rule is available: deem the gap decisive when it exceeds τ_{det} , on the grounds that the detector then had time to fire inside the very interval τ_{ARF} elides, so the substitution changes the outcome of the race. That is a far more meaningful yardstick than a fraction of an arbitrary horizon, and we record it as such.

It cannot serve here, because τ_{det} depends on λ . At $\lambda = 50$, where 1999 runs out of 2000 never alarm, τ_{det} is infinite and the criterion is never met: τ_{ARF} would be declared acceptable everywhere. At $\lambda = 8$ it is met almost everywhere. The same indicator would be valid or invalid according to how the monitor was tuned — whereas construct validity is a property of the indicator, not of what it is raced against. The rule belongs to the operational question of whether the gap matters, not to the present one of whether τ_{ARF} measures what it claims.

With this criterion, on the full grid: for $q = 25\%$, the gap falls below 100 steps as soon as $\Delta e \gtrsim 0.24$ (median gap of 291 at $\Delta e = 0.028$, versus 6.5–7 steps beyond $\Delta e = 0.48$) — τ_{ARF} becomes an acceptable substitute again there. For $q = 50\%$, the same threshold is only crossed around $\Delta e \gtrsim 0.33$ (gap of 633 at $\Delta e = 0.028$, versus 18–19 steps at the top of the grid). For $q = 75\%$, the gap *never* falls below 100 steps anywhere on the tested grid. Across the 18 amplitudes where fewer than half of the runs are censored for $q = 75\%$, the gap runs from 167 to 1368 steps, i.e. 6.0 to 14.3 times τ_{ARF} itself. The two censored amplitudes are excluded here, and in particular the apparent maximum of 1393 at $\Delta e = 0.141$, which the censoring rate of 54% makes unusable. τ_{ARF} is at no point an acceptable proxy for “three-quarters of the forest have been renewed”.

3 The Conjecture $\tau_{\text{ARF}} \leq \tau_{\text{err}}(\rho)$

Where $\tau_{\text{swap}}(q)$ dates recovery by counting trees, $\tau_{\text{err}}(\rho)$ dates it by watching the error itself come back down:

$$\tau_{\text{err}}(\rho) = \inf\{t > \tau^* : \bar{e}_t - \hat{p}_0 \leq \rho \Delta e\} - \tau^*, \quad (2)$$

the first instant at which the excess error has fallen to a fraction ρ of the jump it suffered at the drift. We take $\rho = 0.25$ throughout — three quarters of the drift absorbed. The value is ours to choose, the assignment fixing only $\rho \in (0, 1)$; what it costs at the weakest amplitudes — where $\rho\Delta e$ falls below the noise of $\bar{e}_t$ and the test decides nothing — is set out below.

$e_t \in \{0, 1\}$ is only statistically meaningful once averaged over seeds: on a single run, the sequence is too noisy for a threshold crossing to signal genuine recovery. $\tau_{\text{err}}(\rho)$, which the other sections write τ_{rec} , is therefore computed on the mean curve $\bar{e}_t$ alone — as $\bar{e}_t$ in (2) already indicates: one value per amplitude, never one per run.

τ_{ARF} , by contrast, exists on every run, and comparing the two means giving up that resolution. The test is run amplitude by amplitude, between $\tau_{\text{err}}(\rho)$ and the *inter-seed median* of τ_{ARF} — so the conjecture is decided on 20 paired values, not on 2000 runs, and the dispersion of τ_{ARF} within an amplitude plays no part in the verdict.

Conjecture. Incremental learning by the non-replaced Hoeffding trees can bring the ensemble error back down without any tree having changed, particularly at low amplitudes where the shock is too faint to require a replacement. Under this hypothesis, $\tau_{\text{ARF}} \leq \tau_{\text{err}}(\rho)$ would not hold at low Δe .

Test and result. With $\rho = 0.25$, a first threshold crossing ($\bar{e}_t - \hat{p}_0 \leq \rho\Delta e$) with no persistence condition shows 4 apparent violations, at the four lowest amplitudes of the grid ($\Delta e = 0.028, 0.085, 0.141$ and 0.194). They fall in two distinct ways, and the distinction matters. Figure 3 shows the whole test at a glance: any point falling below the median- τ_{ARF} curve is a violation.

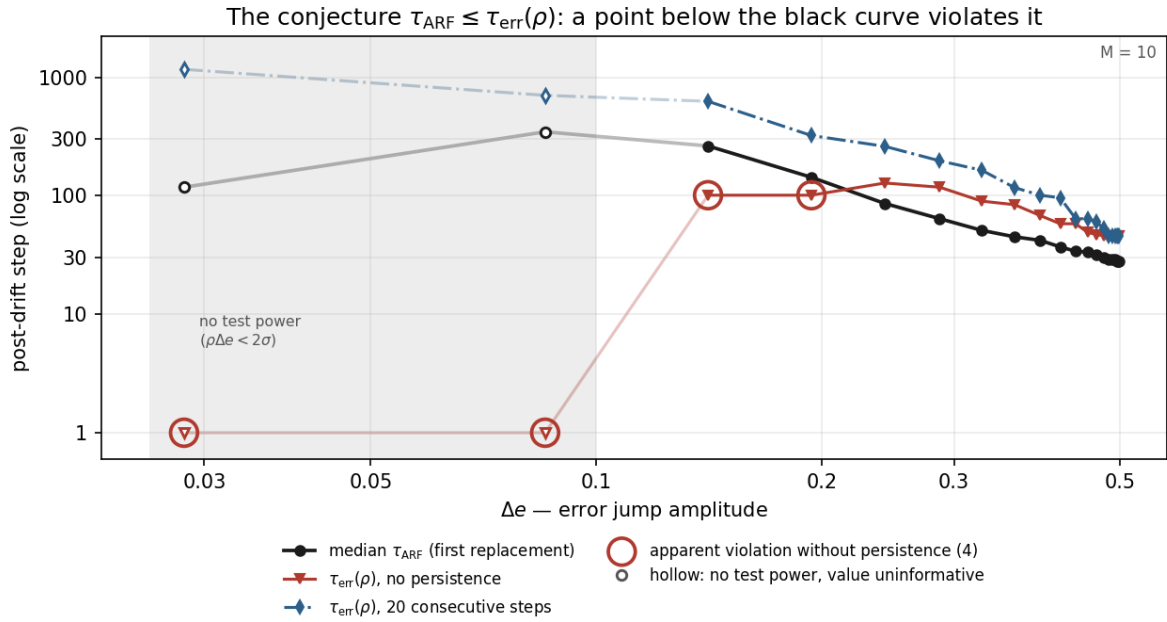


Figure 3: The conjecture tested directly, on all 20 amplitudes: median τ_{ARF} against $\tau_{\text{err}}(\rho)$ under both crossing rules. A point below the black curve violates $\tau_{\text{ARF}} \leq \tau_{\text{err}}(\rho)$. Without a persistence requirement there are four, all at the low end of the grid; requiring the condition to hold for 20 consecutive steps removes both of the two that occur where the test can decide. Hollow markers, inside the shaded band, are the two amplitudes where $\rho\Delta e < 2\sigma$: the persistence rule still returns a value there, and that value still sits above τ_{ARF} , but a threshold buried in the noise dates the noise rather than the recovery. Those two points are not a verdict either way, which is why the count of remaining violations is taken over 18 amplitudes and not 20.

Two of them are artefacts of a powerless test. The residual noise of the mean curve (σ , estimated over the last 500 steps) is amplitude-dependent, ranging from 0.0154 to 0.0047 across the grid and broadly decreasing with amplitude; at the two lowest amplitudes it is of the same order as the threshold $\rho\Delta e$ itself ($\rho\Delta e = 0.007$ against $\sigma = 0.0153$, and 0.021 against 0.0149), so that $\rho\Delta e < 2\sigma$ and the test can decide nothing. There, the crossing is detected at the very first post-drift step, which is a fluctuation and not a recovery. Figure 4 locates the boundary: the threshold $\rho\Delta e$ grows linearly with Δe while the noise floor 2σ stays roughly flat, and the two curves cross between $\Delta e = 0.085$ and

$\Delta e = 0.141$. Below that crossing the test has no power; above it, it regains power — which is precisely why the two remaining violations cannot be disposed of in the same way.

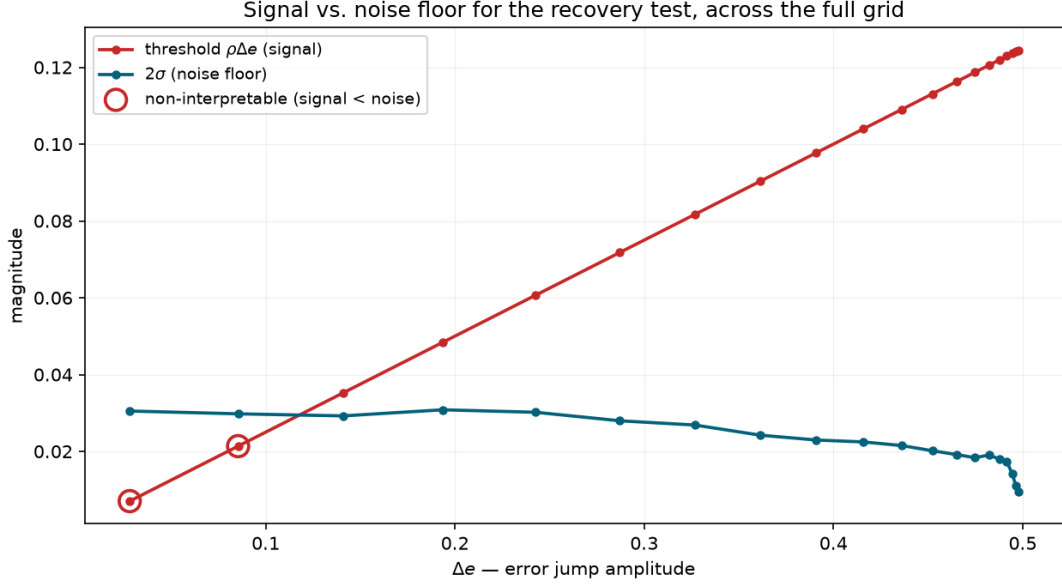


Figure 4: Threshold $\rho\Delta e$ (signal) versus 2σ (noise floor) across all 20 amplitudes. The two curves cross between $\Delta e = 0.085$ and $\Delta e = 0.141$, which separates the two non-interpretable amplitudes (circled) from the rest of the grid. This boundary is a statement about the power of the test, not about where the violations are: two of the four violations sit above it.

The other two, at $\Delta e = 0.141$ and 0.194 , occur where the test *is* interpretable ($\rho\Delta e > 2\sigma$), and they cannot be dismissed on power grounds. They are eliminated by the persistence requirement instead: the crossing is recorded at step 101 in both cases, ahead of the median τ_{ARF} (263 and 143 respectively), but it is not sustained. Requiring the condition to hold for 20 consecutive steps rather than one, the crossing moves to steps 626 and 322, and all four violations disappear: **0 violations across the 18 interpretable amplitudes**. The persistence condition therefore does real work here. It is not merely a cosmetic filter on the two amplitudes where the test was already blind.

Conclusion. On the 18 amplitudes where the test has power, $\tau_{\text{ARF}} \leq \tau_{\text{err}}(\rho)$ holds without exception: the first replacement always precedes the actual recovery of the error. That is the optimistic bias of (1) again, now against a second yardstick.

The conjecture is therefore not confirmed, and at the two weakest amplitudes it is not refuted either — the test cannot decide there. Settling them would take more seeds, and the $1/\sqrt{n}$ law makes that expensive: 194 at $\Delta e = 0.085$, but 1 874 at 0.028, and more in practice, a crossing instant being harder to locate where the curve is flattest.

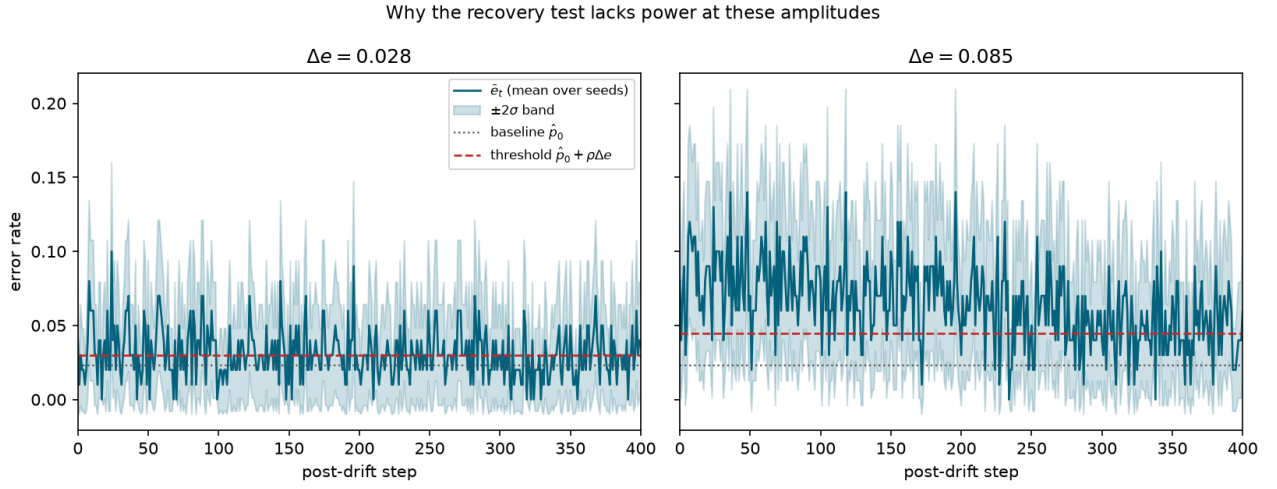


Figure 5: Mean post-drift error curve $\bar{e}_t$ (100 seeds) with a $\pm 2\sigma$ band, where σ is the per-step binomial standard error $\sqrt{\bar{e}_t(1 - \bar{e}_t)/n}$, against the baseline $\hat{p}_0$ and the recovery threshold $\hat{p}_0 + \rho\Delta e$, at the two amplitudes where the test lacks power ($\rho\Delta e \leq 2\sigma$). The interpretability criterion itself uses a different estimator of the same order, the empirical tail standard deviation of the mean curve over the last 500 steps (0.0153 and 0.0149, against 0.0164 and 0.0239 for the plotted band). The noise band covers the threshold at both amplitudes, visually confirming the lack of statistical power there.